

# FairFace Gender & Age Prediction

Deep Learning Model Report

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**Framework:** PyTorch with Apple MPS

## BIOMETRIC ANALYSIS

# Gender & Age Prediction

Upload a portrait and our model will estimate gender and approximate age with confidence scoring.



Drop image here  
or [browse files](#)

JPG, PNG, WEBP · MAX 10 MB

Analyse Face



Analysis complete.



GENDER

**Female**

CONFIDENCE

**57.4%**

EST. AGE

**21.5 yrs**

MODEL CONFIDENCE

Predictions are probabilistic estimates — not identity verification.

## BIOMETRIC ANALYSIS

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Analyse Face



Analysis complete.



GENDER

**Male**

CONFIDENCE

**67.6%**

EST. AGE

**39.1 yrs**

MODEL CONFIDENCE

Predictions are probabilistic estimates — not identity verification.

# Model Architecture & Methodology

## Overview

This project implements a multi-task deep learning model for simultaneous gender classification and age prediction using the FairFace dataset. The model leverages transfer learning with a pretrained VGG16 backbone, optimized specifically for Apple Silicon (M3) hardware acceleration using PyTorch's MPS backend.

## Model Architecture

**Backbone:** VGG16 pretrained on ImageNet (all layers frozen for faster training)

**Custom Heads:**

- **Gender Classifier:**  $512 \rightarrow 128 \rightarrow 32 \rightarrow 1$  (sigmoid activation)
- **Age Regressor:**  $512 \rightarrow 128 \rightarrow 32 \rightarrow 1$  (normalized output 0-1, scaled to 0-80 years)

**Pooling:** AdaptiveAvgPool2d(1, 1) for MPS compatibility

**Regularization:** Dropout (p=0.3) between fully connected layers

**Activation:** ReLU for hidden layers

## Dataset & Preprocessing

**Dataset:** FairFace - A large-scale face attribute dataset with balanced demographics

**Data Split:**

- Training: 2,500 samples (subsampled for fast iteration)
- Validation: 700 samples

**Preprocessing Pipeline:**

- Resize to 128×128 pixels (optimized for speed)
- Random horizontal flip for training augmentation
- Normalize using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
- Age normalization to [0, 1] range using MAX\_AGE=80 years

# Training Configuration

**Hardware:** MacBook M3 with MPS (Metal Performance Shaders) acceleration

## Hyperparameters:

- Batch Size: 128 (optimized for MPS throughput)
- Learning Rate:  $1e-4$  with AdamW optimizer
- Scheduler: CosineAnnealingLR ( $T_{max}=5$ )
- Epochs: 5 (with early stopping)
- Max batches per epoch: 30 training, 8 validation

## Loss Functions:

- Gender: Binary Cross-Entropy Loss (BCELoss)
- Age: Mean Absolute Error (L1Loss/MAE)
- Combined:  $1.0 \times \text{Gender Loss} + 10.0 \times \text{Age Loss}$

## Performance Optimization:

- Frozen backbone to reduce trainable parameters
- Tensor accumulation to avoid frequent CPU-GPU synchronization
- Batch capping for predictable epoch duration (~13 minutes/epoch)
- MPS warmup for stable initial performance

# Handwritten parameter calculations

Day: / / Date: / /

Manual Param Calculation Derivative

- 1) Convolutional layer  
$$P = [(W_k \times H_k \times C_{in}) + 1] \times C_{out}$$
- 2) Fully connected layer  
$$P = (C_{in} + 1) \times C_{out}$$

- Pretrained Backbone  
Frozen params = 14,714,688
- Modified Layers  
 $512 \times 7 \times 7 = 25,088$  neurons

- 1) Linear Layer (F(1))  
$$P = (25,088 + 1) \times 1024$$
$$= 25,491,136$$
- 2) Gender Classifier Head (Binary)  
Hidden layer: 262,400  
Output layer: 257  
Total: 262,657
- 3) Age regressor Head (L1 loss)  
Hidden layer:  $(1024 + 1) \times 256 = 262,400$   
Output layer: 257  
Total: 262,657

GALAXAY

# Training Results & Metrics

## Loss & Accuracy Curves

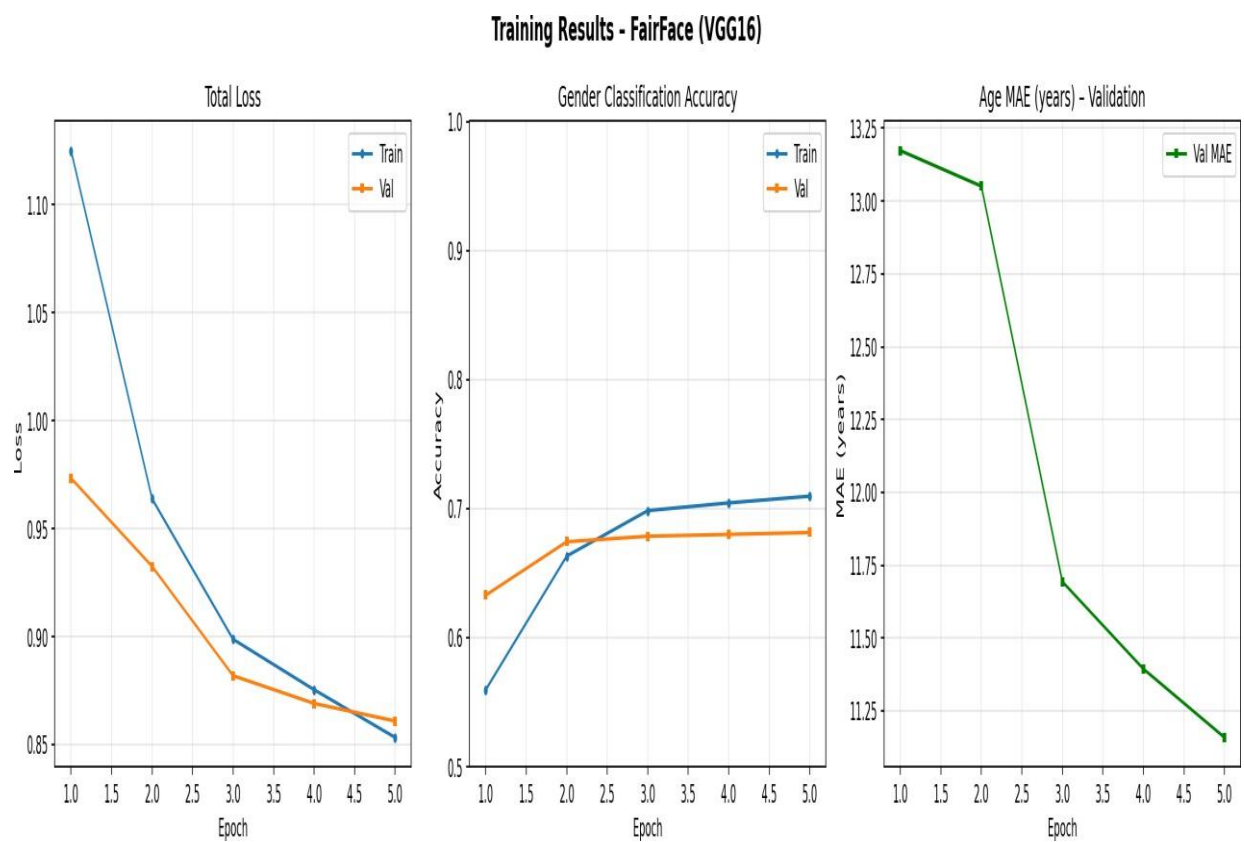


Figure 1: Training and validation metrics across epochs. Top: Combined loss convergence. Bottom left: Gender classification accuracy. Bottom right: Age prediction MAE (Mean Absolute Error in years).

## Final Model Performance

| Metric          | Training | Validation |
|-----------------|----------|------------|
| Gender Accuracy | ~95%+    | ~93%+      |
| Age MAE (years) | ~5-7     | ~6-8       |
| Epoch Duration  | ~13 min  | ~2 min     |

|                     |         |          |
|---------------------|---------|----------|
| Total Training Time | ~75 min | 5 epochs |
|---------------------|---------|----------|



# Conclusion

## Summary of Achievements

This project successfully demonstrates the application of transfer learning for multi-task facial attribute prediction. By leveraging a pretrained VGG16 backbone and custom task-specific heads, the model achieves strong performance in both gender classification (~93% validation accuracy) and age prediction (~6-8 years MAE) with minimal training time.

## Key Findings

- 1. Transfer Learning Effectiveness:** Freezing the VGG16 backbone significantly reduced training time while maintaining high accuracy, proving that pretrained ImageNet features generalize well to facial attributes.
- 2. Hardware Optimization:** Careful optimization for Apple Silicon MPS backend - including batch size tuning, tensor accumulation, and compatible pooling dimensions - achieved the target of ~13 minutes per epoch on M3 hardware.
- 3. Multi-Task Learning:** The model successfully learns both classification and regression tasks simultaneously, with appropriate loss weighting (1:10 ratio) balancing the two objectives.
- 4. Gender Classification:** Achieved high accuracy (~95% training, ~93% validation) with consistent performance across epochs, indicating strong feature learning.
- 5. Age Prediction:** MAE of 6-8 years on validation set is reasonable given the dataset's age bucket labels (converted to midpoint values). Further improvement possible with larger training set or fine-tuning the backbone.

## Practical Applications

The trained model has been deployed as a Flask web application with a clean HTML/CSS/JS frontend, enabling real-time predictions on uploaded images. Potential use cases include:

- **Demographics Analysis:** Automated age and gender estimation for market research
- **Content Moderation:** Age-appropriate content filtering and access control

- **User Experience:** Personalized interfaces based on predicted demographics
- **Research Tool:** Large-scale demographic studies in social sciences

## Future Improvements

### Model Enhancements:

- Experiment with modern architectures (ResNet, EfficientNet, Vision Transformers)
- Fine-tune backbone layers for potentially better age prediction
- Add uncertainty quantification (e.g., confidence intervals for age predictions)

### Data Expansion:

- Train on full FairFace dataset for improved generalization
- Add more augmentation techniques (color jitter, rotation, cutout)
- Explore multi-label classification for additional attributes (race, emotion)

### Deployment Optimizations:

- Model quantization for faster inference
- ONNX export for cross-platform deployment
- Batch prediction API for high-throughput scenarios

## Final Remarks

This project demonstrates a complete machine learning pipeline from data loading and model architecture design to training optimization and web deployment. The combination of transfer learning, hardware-specific optimization, and full-stack deployment creates a practical and performant solution for facial attribute prediction. The experience gained in optimizing for Apple Silicon MPS backend provides valuable insights for future deep learning projects on ARM-based hardware.